PS#3: Integration by substitution – Answer key QR

1. (Exercise 5.3.6.11)
   (a) $-\ln(\cos(x))+C$
   (b) $\ln(\sin(x))+C$
   (c) $\ln(\sec(x)+\tan(x))+C$
   (d) $\int \sec(x)\,dx$
   (e) $\int \sec(x)\,dx = \ln(\sec(x)+\tan(x))+C$
   (f) $\int \csc(x)\,dx = -\ln(\cot(x)+\csc(x))+C$
2. (Exercise 5.3.6.12)
   (a) $u = x-1$; $x = u-1$; $du = dx$
   (b) $\int (u+1)\cdot u^{1/2}\,du = \int u^{3/2}+u^{1/2}\,du$
   (c) $\frac{2}{5}(x-1)^{5/2}+\frac{2}{3}(x-1)^{3/2}+C$
   (d) $\frac{2}{7}(x-1)^{7/2}+\frac{4}{5}(x-1)^{5/2}+\frac{2}{3}(x-1)^{3/2}+C$; $\frac{1}{3}(x^2+1)^{3/2}+C$
3. (Exercise 5.3.6.13)
   (a) no way to make du
   (b) $\int \sin(x)\cdot(1-\cos^2(x))\,dx$
   (c) yay du works now
   (d) $-\cos(x)+\frac{1}{3}\cos^3(x)+C$
   (e) $\sin(x)-\frac{1}{3}\sin^3(x)+C$
4. (Exercise 5.3.6.14)
   - Part (c): 95.7809 mWh
   - Part (d): 3.99087 mWh